

$$\begin{array}{ccc}
 A_k & \xrightarrow{i_k} & B_k \\
 a_k \downarrow & & \downarrow b_k \\
 X_{k-1} & \xrightarrow{j_k} & P_k = X_{k-1} \sqcup_{A_k} B_k
 \end{array}$$